

Mathematical Applications

In previous chapters, many references have been made to the possibility of applying mathematical techniques to option strategies. Those techniques are developed in this chapter. Although the average investor—public, institutional, or floor trader—normally has a limited grasp of advanced mathematics, the information in this chapter should still prove useful. It will allow the investor to see what sorts of strategy decisions could be aided by the use of mathematics. It will allow the investor to evaluate techniques of an information service. Additionally, if the investor is contemplating hiring someone knowledgeable in mathematics to do work for him, the information to be presented may be useful as a focal point for the work. The investor who does have a knowledge of mathematics and also has access to a computer will be able to directly use the techniques in this chapter.

THE BLACK-SCHOLES MODEL

Since an option's price is the function of stock price, striking price, volatility, time to expiration, and short-term interest rates, *it is logical that a formula could be drawn up to calculate option prices* from these variables. Many models have been conceived since listed options began trading in 1973. Many of these have been attempts to improve on one of the first models introduced, the *Black-Scholes model*. This model was introduced in early 1973, very near the time when listed options began trading. It was made public at that time and, as a result, gained a rather large number of adherents. The formula is rather easy to use in that the equations are short and the number of variables is small.

The actual formula is:

$$\text{Theoretical option price} = pN(d_1) - se^{-rt}N(d_2)$$

$$\text{where } d_1 = \frac{\ln\left(\frac{p}{s}\right) + \left(r + \frac{v^2}{2}\right)t}{v\sqrt{t}}$$

$$d_2 = d_1 - v\sqrt{t}$$

The variables are:

p = stock price

s = striking price

t = time remaining until expiration, expressed as a percent of a year

r = current risk-free interest rate

v = volatility measured by annual standard deviation

$\ln$ = natural logarithm

$N(x)$ = cumulative normal density function

*An important by-product of the model is the exact calculation of the delta—that is, the amount by which the option price can be expected to change for a small change in the stock price. The delta was described in Chapter 3 on call buying, and is more formally known as the *hedge ratio*.*

$$\text{Delta} = N(d_1)$$

The formula is so simple to use that it can fit quite easily on most programmable calculators. In fact, some of these calculators can be observed on the exchange floors as the more theoretical floor traders attempt to monitor the present value of option premiums. Of course, a computer can handle the calculations easily and with great speed. A large number of Black-Scholes computations can be performed in a very short period of time.

The cumulative normal distribution function can be found in tabular form in most statistical books. However, for computation purposes, it would be wasteful to repeatedly look up values in a table. Since the normal curve is a smooth curve (it is the “bell-shaped” curve used most commonly to describe population distributions), the cumulative distribution can be approximated by a formula:

$$x = 1 - z(1.330274y^5 - 1.821256y^4 + 1.781478y^3 - .356538y^2 + .3193815y)$$

$$\text{where } y = \frac{1}{1 + .2316419|\sigma|} \quad \text{and} \quad z = .3989423e^{-\sigma^2/2}$$

$$\text{Then } N(\sigma) = x \text{ if } \sigma > 0 \quad \text{or} \quad N(\sigma) = 1 - x \text{ if } \sigma < 0$$

This approximation is quite accurate for option pricing purposes, since one is not really interested in thousandths of a point where option prices are concerned.

Example: Suppose that XYZ is trading at 45 and we are interested in evaluating the July 50 call, which has 60 days remaining until expiration. Furthermore, assume that the volatility of XYZ is 30% and that the risk-free interest rate is currently 10%. The theoretical value calculation is shown in detail, in order that those readers who wish to program the model will have something to compare their calculations against.

Initially, determine t , d_1 , and d_2 , by referring to the formulae on the previous page:

$$\begin{aligned} t &= 60/365 = .16438 \text{ years} \\ d_1 &= \frac{\ln(45/50) + (.1 + .3 \times .3/2) \times .16438}{.3 \times \sqrt{.16438}} \\ &= \frac{-.10536 + (.145 \times .16438)}{.3 \times .40544} = -.67025 \\ d_2 &= -.67025 - .3\sqrt{.16438} = -.67025 - (.3 \times .40544) = -.79189 \end{aligned}$$

Now calculate the cumulative normal distribution function for d_1 and d_2 by referring to the above formulae:

$$\begin{aligned} d_1 &= -.67025 \\ y &= \frac{1}{1 + (.2316419) | - .67025 |} = \frac{1}{1.15526} = .86561 \\ z &= .3989423e^{-(.67025)^2/2} \\ &= .3989423e^{-0.22462} = .31868 \end{aligned}$$

There are too many calculations involved in the computation of the fifth-order polynomial to display them here. Only the result is given:

$$x = .74865$$

Since we are determining the cumulative normal distribution of a negative number, the distribution is determined by subtracting x from 1.

$$N(d_1) = N(-.67025) = 1 - x = 1 - .74865 = .25134$$

In a similar manner, which requires computing new values for x , y , and z ,

$$N(d_2) = N(-.79189) = 1 - .78579 = .21421$$

Now, returning to the formula for theoretical option price, we can complete the calculation of the July 50 call's theoretical value, called value here for short:

$$\begin{aligned}\text{value} &= 45 \times N(d_1) - 50 \times e^{-.1 \times .16438} \times N(d_2) \\ &= 45 \times .25134 - 50 \times .9837 \times .21421 \\ &= .7746\end{aligned}$$

Thus, the theoretical value of the July 50 call is just slightly over $\frac{3}{4}$ of a point. Note that the delta of the call was calculated along the way as $N(d_1)$ and is equal to just over .25. That is, the July 50 call will change price about $\frac{1}{4}$ as fast as the stock for a small price change by the stock.

This example should answer many of the questions that readers of the previous editions have posed. The reader interested in a more in-depth description of the model, possibly including the actual derivation, should refer to the article "Fact and Fantasy in the Use of Options."¹ One of the less obvious relationships in the model is that call option prices will increase (and put option prices will decrease) as the risk-free interest rate increases. It may also be observed that the model correctly preserves relationships such as increased volatility, higher stock prices, or more time to expiration, which all imply higher option prices.

CHARACTERISTICS OF THE MODEL

Several aspects of this model are worth further discussion. First, the reader will notice that *the model does not include dividends paid by the common stock*. As has been demonstrated, dividends act as a negative effect on call prices. Thus, direct application of the model will tend to give inflated call prices, especially on stocks that pay relatively large dividends. There are ways of handling this. Fisher Black, one of the coauthors of the model, suggested the following method: Adjust the stock price to be used in the formula by subtracting, from the current stock price, the present worth of the dividends likely to be paid before maturity. Then calculate the option price. Second, assume that the option expires just prior to the last ex-dividend date preceding actual option expiration. Again adjust the stock price and calculate the option price. Use the higher of the two option prices calculated as the theoretical price.

Another, less exact, method is to apply a weighting factor to call prices. The weighting factor would be based on the dividend payment, with a heavier weight being applied to call options on high-yielding stock. It should be pointed out that, in many of the applications that are going to be prescribed, it is not necessary to know the exact theoretical price

¹Fisher Black, *Financial Analysts Journal*, July–August 1975, pp. 36–70.

of the call. Therefore, the dividend “correction” might not have to be applied for certain strategy decisions.

The model is based on a lognormal distribution of stock prices. Even though the normal distribution is part of the model, the inclusion of the exponential functions makes the distribution lognormal. For those less familiar with statistics, a normal distribution has a bell-shaped curve. This is the most familiar mathematical distribution. The problem with using a normal distribution is that it allows for negative stock prices, an impossible occurrence. Therefore, the lognormal distribution is generally used for stock prices, because it implies that the stock price can have a range only between zero and infinity. Furthermore, the upward (bullish) bias of the lognormal distribution appears to be logically correct, since a stock can drop only 100% but can rise in price by more than 100%. Many option pricing models that antedate the Black–Scholes model have attempted to use empirical distributions. An empirical distribution has a different shape than either the normal or the lognormal distribution. Reasonable empirical distributions for stock prices do not differ tremendously from the lognormal distribution, although they often assume that a stock has a greater probability of remaining stable than does the lognormal distribution. Critics of the Black–Scholes model claim that, largely because it uses the lognormal distribution, the model tends to overprice in-the-money calls and underprice out-of-the-money calls. This criticism is true in some cases, but does not materially subtract from many applications of the model in strategy decisions. True, if one is going to buy or sell calls solely on the basis of their computed value, this would create a large problem. However, if strategy decisions are to be made based on other factors that outweigh the overpriced/underpriced criteria, small differentials will not matter.

The computation of volatility is always a difficult problem for mathematical application. In the Black–Scholes model, volatility is defined as the annual standard deviation of the stock price. This is the regular statistical definition of standard deviation:

$$\sigma^2 = \frac{\sum_{i=1}^n (P_i - P)^2}{n - 1}$$

$$v = \sigma / P$$

where

P = average stock price of all P_i 's

P_i = daily stock price

n = number of days observed

v = volatility

When volatility is computed using past stock prices, it is called a historical volatility. *The volatilities of stocks tend to change over time.* Certain predictable factors, such as a

large stock split increasing the float of the stock, can reduce the volatility. The entry of a company into a more speculative area of business may increase the volatility. Other, less well-defined factors can alter the volatility as well. Since the volatility is a very crucial element of the pricing model, it is important that the modeler use a reasonable estimate of the current volatility. *It has become apparent that an annual standard deviation is not accurate, because it encompasses too long a period of time.* Recent efforts by many modelers have suggested that one should perhaps weight the recent stock price action more heavily than older price action in arriving at a current volatility. This is a possible approach, but the computation of such factors may introduce as much error as using the annual standard deviation does. The problem of accurately computing the volatility is critical, because the model is so sensitive to it.

Computing Lognormal Historical Volatility. The above calculation does not give the proper input for the Black–Scholes model because the model assumes that the *logarithms* of changes in price are normally distributed, not the prices themselves. That is, the term P_i in the above formula should be changed.

Example: XYZ closed at 51 today and at 50 yesterday. Thus, its percentage change for the day is $51/50 = 1.02$. The natural logarithm of 1.02 is then based on the volatility formula:

$$\ln(51/50) = \ln(1.02) = 0.0198$$

This is similar to saying that arithmetically the stock was up 2% today, but on a log-normal basis, it was only up 1.98%

If the stock is down, this method will yield a negative number. Suppose that on the following day, XYZ declined from 51 back to 50. The number to use in the volatility formula would then be:

$$\ln(50/51) = \ln(0.9804) = -0.0198$$

A new equation can now be formulated using this concept. It will yield volatilities that are consistent with the Black–Scholes model:

$$v = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

where $X_i = \ln(P_i/P_{i-1})$; P_i = closing price on day i and $\bar{X}$ = the average of the X_i 's over the desired number of days.

So to compute a 10-day historical volatility, one would need 11 observations. In the following example, do not be concerned with the complete details if you do not plan to compute the volatilities yourself; they are provided for the mathematician or programmer who needs to check his work:

Day	XYZ Stock	P_i/P_{i-1}	$X_i = \ln(P_i/P_{i-1})$	$(X_i - \bar{X})^2$
1	153.875			
2	153.625	.9984	-.0016	.000020
3	151	.9829	-.0172	.000405
4	146	.9669	-.0337	.001336
5	144.125	.9872	-.0129	.000250
6	147.25	1.0217	.0215	.000345
7	146.25	.9932	-.0068	.000094
8	149.5	1.0222	.0220	.000365
9	152.5	1.0201	.0199	.000289
10	158.625	1.0402	.0394	.001332
11	158.375	0.9984	-.0016	.000020
AVG: 0.0028825				Σ : 0.004455

The average of the lns (4th column) over the 10 days is 0.00288.

The difference of each ln from the mean, squared, is then summed (5th column). For example, for day 1 the term is $(-.0016 - .00288)^2 = .00002$. This is the top number in the far right-hand column. This process can be computed for each number in the “ln” column. The sum of all these terms is 0.004455.

$$\text{Now } v = \sqrt{(.004455/9)} = 0.022249$$

This is a 10-day volatility. To convert it into an annual volatility, we need to multiply by the square root of the number of trading days in a year. Since there are approximately 260 trading days in a year, the final volatility would be:

$$v = 0.022249 \times \sqrt{(260)} = 0.3587$$

Thus, one could say that the volatility of XYZ is 36% on an annualized basis.

This is then the proper way to calculate historical volatility. Obviously, the strategist can calculate 10-, 20-, and 50-day and annual volatilities if he wishes—or any other number for that matter. In certain cases, one can discern valuable information about a stock or future and its options by seeing how the various volatilities compare with one another.

COMPUTING COMPOSITE IMPLIED VOLATILITY

There is, in fact, a way in which *the strategist can let the market compute the volatility for him*. This is called using the *implied volatility*; that is, the volatility that the market itself is implying. This concept makes the assumption that, for options with striking prices close to the current stock price and for options with relatively large trading volume, the market is fairly priced. This is something like an efficient market hypothesis. If there is enough trading interest in an option that is close to the money, that option will generally be fairly priced. Once this assumption has been made, a corollary arises: *If the actual price of an option is the fair price, it can be fixed in the Black–Scholes equation while letting volatility be the unknown variable*. The volatility can be determined by iteration. In fact, this process of iterating to compute the volatility can be done for each option on a particular underlying stock. This might result in several different volatilities for the stock. If one weights these various results by volume of trading and by distance in- or out-of-the-money, a single volatility can be derived for the underlying stock. This volatility is based on the closing price of all the options on the underlying stock for that given day.

Example: XYZ is at 33 and the closing prices are given in Table 28-1. Each option has a different implied volatility, as computed by determining what volatility in the Black–Scholes model would result in the closing price for each option: That is, if .34 were used as the volatility, the model would give $4\frac{1}{2}$ as the price of the January 30 call. In order to rationally combine these volatilities, weighting factors must be applied before a volatility for XYZ stock itself can be arrived at.

The weighting factors for volume are easy to compute. The factor for each option is merely that option's daily volume divided by the total option volume on all XYZ options (Table 28-2). The weighting functions for distance from the striking price should probably not be linear. For example, if one option is 2 points out-of-the-money and another is 4 points out-of-the-money, the former option should not necessarily get twice as much weight as the latter. Once an option is too far in- or out-of-the-money, it should not be given much or any weight at all, regardless of its trading volume. Any parabolic function of the following form should suffice:

$$\text{Weighting factor} \begin{cases} = \frac{(x-a)^2}{a^2} & \text{if } x \text{ is less than } a \\ = 0 & \text{if } x \text{ is greater than } a \end{cases}$$

TABLE 28-1.
Implied volatilities, closing price, and volume.

Option	Option Price	Volume	Implied Volatility
January 30	4.50	50	.34
January 35	1.50	90	.28
April 35	2.50	55	.30
April 40	1.50	5	.38
		<u>200</u>	

TABLE 28-2.
Volume weighting factors.

Option	Volume	Volume Weighting Factor
January 30	50	.25 (50/200)
January 35	90	.45 (90/200)
April 35	55	.275 (55/200)
April 40	5	.025 (5/200)

where x is the percentage distance between stock price and strike price and a is the maximum percentage distance at which the modeler wants to give any weight at all to the option's implied volatility.

Example: An investor decides that he wants to discard options from the weighting criterion that have striking prices more than 25% from the current stock price. The variable, a , would then be equal to .25. The weighting factors, with XYZ at 33, could thus be computed as shown in Table 28-3. To combine the weighting factors for both volume and distance from strike, the two factors are multiplied by the implied volatility for that option. These products are summed up for all the options in question. This sum is then divided by the products of the weighting factors, summed over all the options in question. As a formula, this would read:

$$\text{Implied volatility} = \frac{\sum (\text{Volume factor} \times \text{Distance factor} \times \text{Implied volatility})}{\sum (\text{Volume factor} \times \text{Distance factor})}$$

In our example, this would give an implied volatility for XYZ stock of 29.8% (Table 28-4). Note that the implied volatility, .298, is not equal to any of the individual option's implied volatilities. Rather, it is a composite figure that gives the most weight to the heavily

TABLE 28-3.
Distance weighting factors.

Option	Distance from Stock Price	Distance Weighting Factor
January 30	.091 (3/33)	.41
January 35	.061 (2/33)	.57
April 35	.061 (2/33)	.57
April 40	.212 (7/33)	.02

TABLE 28-4.
Option's implied volatility.

Option	Volume Factor	Distance Factor	Option's Implied Volatility
January 30	.25	.41	.34
January 35	.45	.57	.28
April 35	.275	.57	.30
April 40	.025	.02	.38
$\text{Implied volatility} = \frac{.25 \times .41 \times .34 + .45 \times .57 \times .28 + .275 \times .57 \times .30 + .025 \times .02 \times .38}{.25 \times .41 + .45 \times .57 + .275 \times .57 + .025 \times .02}$ $= .298$			

traded, near-the-money options, and very little weight to the lightly traded (5 contracts), deeply out-of-the-money April 40 call. This implied volatility is still a form of standard deviation, and can thus be used whenever a standard deviation volatility is called for.

This method of computing volatility is quite accurate and proves to be sensitive to changes in the volatility of a stock. For example, as markets become bullish or bearish (generating large rallies or declines), most stocks will react in a volatile manner as well. Option premiums expand rather quickly, and this method of implied volatility is able to pick up the change quickly. One last bit of fine-tuning needs to be done before the final volatility of the stock is arrived at. On a day-to-day basis, the implied volatility for a stock—especially one whose options are not too active—may fluctuate more than the strategist would like. A smoothing effect can be obtained by taking a moving average of the last 20 or 30 days' implied volatilities. An alternative that does not require the saving

of many previous days' worth of data is to use a momentum calculation on the implied volatility. For example, today's final volatility might be computed by adding 5% of today's implied volatility to 95% of yesterday's final volatility. This method requires saving only one previous piece of data—yesterday's final volatility—and still preserves a "smoothing" effect.

Once this implied volatility has been computed, it can then be used in the Black–Scholes model (or any other model) as the volatility variable. Thus one could compute the theoretical value of each option according to the Black–Scholes formula, utilizing the implied volatility for the stock. Since the implied volatility for the stock will most likely be somewhat different from the implied volatility of this particular option, there will be a discrepancy between the option's actual closing price and the theoretical price as computed by the model. This differential represents the amount by which the option is theoretically overpriced or underpriced, *compared to other options on that same stock.*

Computing a Volatility Skew

There is not a single, definitive way to calculate a single number for each stock each day that represents the skew in the options, but this is one acceptable way. Essentially, the process is this:

1. Calculate the individual implied volatility of each option.
2. Calculate the standard deviation of the series in step 1. It is not necessary to weight these individual implied volatilities as one does in the composite implied volatility calculation. Rather, merely compute the standard deviation of the set of implied volatilities. Also, note that one may want to eliminate options that are essentially trading with little or no time value premium from this standard deviation calculation, since they are not representative of the "normal" options on this stock.
3. Divide the result of step 2 by the composite implied volatility, computed as shown in the preceding section.

Example: XYZ is trading at 6.50. It has several listed options, with various individual implied volatilities.

Option	Implied Volatility
Mar 5 call	85.0%
June 5 call	77.5%
Mar 7.5 call	75.0%
June 7.5 call	70.0%

The standard deviation of these four numbers is 6.25. Note that this number does not take into account the price or the volume of the individual options. However, deeply in- or out-of-the-money options would not be included if their time value premium is extremely small.

Furthermore, assuming that the composite implied volatility of the above four options (which *does* use volume and distance in- or out-of-the-money), is 75.0%, the “skew factor” for this stock on this day would be:

$$\text{Skew factor} = 6.25 / 75.0 = 8.3\%$$

Similar skew factors would be computed for all stocks, and then ranked. Those with the highest skew factors are likely to have a distinct volatility skew. One would have to look at the implied volatilities of the individual options on any particular stock with a large skew factor to see what is causing the skew.

If an “event” (FDA hearing, lawsuit verdict, earnings report, etc.) is due, that might cause a horizontal skew—where the implied volatilities of the options expiring just after the anticipated event are more expensive than all other options. Conversely, in a bearish market, there might be a vertical skew, where options at lower strikes, for example, have higher implied volatilities than options at higher strikes. Strategies dealing with these skews will be discussed in the chapter on Volatility Trading Techniques.

Once the Composite Implied Volatility and the Volatility Skew Factor are computed, one should consider keeping a database of daily values for every stock, index, ETF, and futures contract. With this information, one would then be able to compute percentiles of implied volatility and skew, looking back over time. These are useful statistics to help one decide if a particular stock’s options are indeed expensive or cheap, or if they are unusually skewed.

EXPECTED RETURN

Certain investors will enter positions only when the historical percentages are on their side. When one enters into a transaction, he normally has a belief as to the possibility of making a profit. For example, when he buys stock he may think that there is a “good chance” that there will be a rally or that earnings will increase. The investor may consciously or unconsciously evaluate the probabilities, but invariably, an investment is made based on a positive expectation of profit. Since options have fixed terms, they lend themselves to a more rigorous computation of expected profit than the aforementioned intuitive appraisal. This more rigorous approach consists of computing the expected return. *The expected return is nothing more than the return that the position should yield over a large number of cases.*

TABLE 28-5.
Calculation of expected returns.

Price of XYZ in 6 Months	Chance of XYZ Being at That Price
Below 30	20%
31	10%
32	10%
33	10%
34	10%
Above 35	40%
	100%

A simple example may help to explain the concept. The crucial variable in computing expected return is to outline what the chances are of the stock being at a certain price at some future time.

Example: XYZ is selling at 33, and an investor is interested in determining where XYZ will be in 6 months. Assume that there is a 20% chance of XYZ being below 30 in 6 months, and that there is a 40% chance that XYZ will be above 35 in 6 months. Finally, assume that XYZ has an equal 10% chance of being at 31, 32, 33, or 34 in 6 months. All other prices are ignored for simplification. Table 28-5 summarizes these assumptions.

Since the percentages total 100%, all the outcomes have theoretically been allowed for. Now suppose a February 30 call is trading at 4 and a February 35 call is trading at 2 points. A bull spread could be established by buying the February 30 and selling the February 35. This position would cost 2 points—that is, it is a 2-point debit. The spreader could make 3 points if XYZ were above 35 at expiration for a return of 150%, or he could lose 100% if XYZ were below 30 at expiration. The expected return for this spread can be computed by multiplying the outcome at expiration for each price by the probability of being at that price, and then summing the results. For example, if XYZ is below 30 at expiration, the spreader loses \$200. It was assumed that there is a 20% chance of XYZ being below 30 at expiration, so the expected loss is 20% times \$200, or \$40. Table 28-6 shows the computation of the expected results at all the prices. The total expected profit is \$100. This means that the expected return (profit divided by investment) is 50% (\$100/\$200). This appears to be an attractive spread, because the spreader could “expect” to make 50% on his money, less commissions.

What has really been calculated in this example is merely *the return that one would expect to make in the long run if he invested in the same position many times throughout history*. Saying that a particular position has an expected return of 8 or 9% is no different

TABLE 28-6.
Computation of expected profit.

XYZ Price at Expiration	Chance of Being at That Price (A)	Profit at That Price (B)	Expected Profit: (A) x (B)
Below 30	20%	-\$200	-\$ 40
31	10%	- 100	- 10
32	10%	0	0
33	10%	+ 100	+ 10
34	10%	+ 200	+ 20
Above 35	40%	+ 300	+ 120
		Total expected profit	\$100

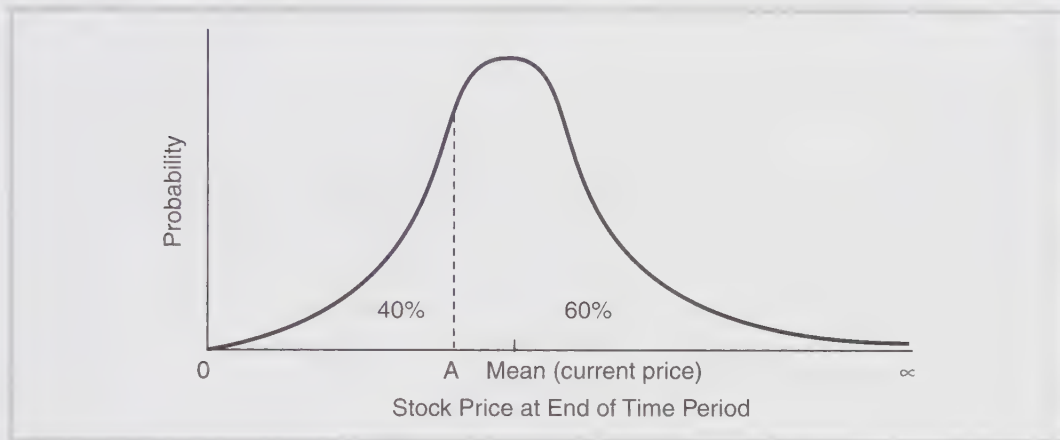
from saying that common stocks return 8 or 9% in the long run. Of course, in bull markets stock would do much better, and in bear markets much worse. In a similar manner, this example bull spread with an expected return of 50% may do as well as the maximum profit or as poorly as losing 100% in any one case. It is the total return on many cases that has the expected return of 50%. *Mathematical theory holds that, if one constantly invests in positions with positive expected returns, he should have a better chance of making money.*

As is readily observable, the selection of what percentages to assign to the possible outcomes in the stock price is a crucial choice. In the example above, if one altered his assumption slightly so that XYZ had a 30% chance of being below 30 and a 30% chance of being above 35 at expiration, the expected return would drop considerably, to 25%. Thus, *it is important to have a reasonably accurate and consistent method of assigning these percentages.* Furthermore, the example above was too simplistic, in that it did not allow for the stock to close at any fractional prices, such as 32.50. A correct expected return computation must take into account all possible outcomes for the stock.

Fortunately, there is a straightforward method of computing the expected percentage chance of a given stock being at a certain price at a certain point in time. This computation involves using the distribution of stock prices. As mentioned earlier, the Black-Scholes model assumes a lognormal distribution for stock prices, although many modelers today use nonstandard (empirical or heuristic) distributions. No matter what the distribution, *the area under the distribution curve between any two points gives the probability of being between those two points.*

Figure 28-1 is a graph of a typical lognormal distribution. The peak always lies at the “mean,” or average, of the distribution. For stock price distributions, under the random walk assumption, *the “mean” is generally considered to be the current stock price.* The

FIGURE 28-1.
Typical lognormal distribution.



graph allows one to visualize the probability of being at any given price. Note that there is a fairly great chance that the stock will be relatively unchanged; there is no chance that the stock will be below zero; and there is a bullish bias to the graph—the stock could rise infinitely, although the chances of it doing so are extremely small.

The chance that XYZ will be below the mean at the end of the time period is 50% in a random walk distribution. This also means that 50% of the area under the graph lies to the left of the mean and 50% lies to the right of the mean. Note point A on the graph. Forty percent of the area under the distribution curve lies to the left of point A and 60% lies to the right of it. This means that there is a 40% chance that the stock will be below price A at the end of the time period and a 60% chance that the stock will be above price A. Consequently, the distribution curve can be used to determine the probabilities necessary for the expected return computation. The reader should take note of the fact that these probabilities apply to the *end of the time period*. They say nothing about the chances that XYZ might dip below price A at some time *during* the time period. To compute that percentage, an involved computation is necessary.

The height and width of the distribution graph are determined by the volatility of the underlying stock, when volatility is expressed as a standard deviation. This is consistent with the method of computing volatility described earlier in this chapter. Implied volatility can, of course, be used. Since the option modeler is generally interested in time periods other than one year, *the annual volatility must be converted into a volatility for the time period in question*. This is easily accomplished by the following formula:

$$v_t = v\sqrt{t}$$

where

v = annual volatility

t = time, in years

v_t = volatility for time, t .

As an example, a 3-month volatility would be equal to one-half of the annual volatility. In this case, t would equal .25 (one fourth of a year), so $v_{.25} = v\sqrt{.25} = .5v$.

The necessary groundwork has been laid for the computation of the probability necessary in the expected return calculation. The following formula gives the probability of a stock that is currently at price p being below some other price, q , at the end of the time period. The lognormal distribution is assumed.

Probability of stock being below price q at end of time period t :

$$P(\text{below}) = N\left(\frac{\ln\left(\frac{q}{p}\right)}{v_t}\right)$$

where

N = cumulative normal distribution

p = current price of the stock

q = price in question

$\ln$ = natural logarithm for the time period in question.

If one is interested in computing the probability of the stock being *above* the given price, the formula is

$$P(\text{above}) = 1 - P(\text{below})$$

With this formula, the computation of expected return is quickly accomplished with a computer. One merely has to start at some price—the lower strike in a bull spread, for example—and work his way up to a higher price—the high strike for a bull spread. At each price point in between, the outcome of the spread is multiplied by the probability of being at that price, and a running sum is kept.

Simplistically, the following iterative equation would be used.

$$P(\text{of being at price } x) = P(\text{below } x) - P(\text{below } y)$$

where y is close to but less than x in price. As an example:

$$P \text{ (of being at 32.4)} = P \text{ (below 32.4)} - P \text{ (below 32.3)}$$

Thus, once the low starting point is chosen and the probability of being below that price is determined, one can compute the probability of being at prices that are successively higher merely by iterating with the preceding formula. In reality, *one is using this information to integrate the distribution curve*. Any method of approximating the integral that is used in basic calculus, such as the Trapezoidal Rule or Simpson’s Rule, would be applicable here for more accurate results, if they are desired.

A partial example of an expected return calculation follows.

Example: XYZ is currently at 33 and has an annual volatility of 25%. The previous bull spread is being established—buy the February 30 and sell the February 35 for a 2-point debit—and these are 6-month options. Table 28-7 gives the necessary components for computing the expected return. Column (A), the probability of being below price q , is computed according to the previously given formula, where $p = 33$ and $v_t = .177$ ($= .25\sqrt{1/2}$). The first stock price that needs to be looked at is 30, since all results for the bull spread are equal below that price—a 100% loss on the spread. The calculations would be performed for each tenth of a point up through a price of 35. The expected return is computed by multiplying the two right-hand columns, (B) and (C), and summing the results. Note that column (B) is determined by subtracting successive numbers in column (A). It would not be particularly enlightening to carry this example to completion, since the rest of the computations are similar and there is a large number of them.

TABLE 28-7.
Calculation of expected returns.

Price at Expiration (q)	(A) P (below q)	(B) P (of being at q)	(C) Profit on Spread
30	.295	.295	−\$200
30.10	.301	.006	− 190
30.20	.308	.007	− 180
30.30	.316	.008	− 170
.	.	.	.
.	.	.	.
.	.	.	.

* * *

In theory, if one had the data and the computer power, he could evaluate a wide range of strategies every day and come up with the best positions on an expected return basis. He would probably get a few option buys (puts or calls), some bull spreads, some naked writes and ratio calendar spreads, fewer straddles and ratio writes, and a few covered call writes. This theory would be somewhat difficult to apply in practice, because of the massive numbers of calculations involved and also because of the accuracy of closing price data. It was mentioned previously that a computer will assume that “bad” closing prices are actually attainable. By a “bad” closing price, it is meant that the option did not trade simultaneously with the stock later in the day, and that the actual market for the option is somewhat different in price than is reflected by the closing price for the option. A daily contract volume “screen” will help alleviate this problem. For example, one may want to discard any option from his calculations if that option did not trade a predetermined, minimum number of contracts during the previous day. Data that give closing bids and offers for each option are more expensive but also more reliable, and would alleviate the problem of “bad” closing prices. In addition to a volume screen, another way of reducing the calculations required is to limit oneself to strategies in which one has interest, or which one is reasonably certain will fit in well with his investment objectives. Regardless of the limitations that one places upon the quantity of computations, some computer power is necessary to compute expected return. A sophisticated programmable calculator may be able to provide a real-time calculation, but could never be used to evaluate the entire option universe and come up with a ranking of the preferable situations each day. On-line computer systems are also available that can provide these types of calculations using up-to-the-minute prices. While real-time prices may occasionally be useful, it is not an absolute necessity to have them.

One other by-product of the expected return calculation is that it could be used as another model for predicting the theoretical value of an option. All one would have to do is compute the probabilities of the stock being at each successive price above the striking price of the option by expiration, and sum them up. The result would be the theoretical option value. These data are published by some services and generally give a different theoretical value than would the Black–Scholes model. The reason for the difference most readily lies in the inclusion of the risk-free interest rate in the Black–Scholes model and its omission in the expected return model.

APPLYING THE CALCULATIONS TO STRATEGY DECISIONS

CALL WRITING

One method of ranking covered call writes that was described in Chapter 2 was to rank all the writes that provided at least a minimal acceptable level of return by their probability of not losing money. If one were interested in safety, he might decide to use this approach. Suppose he decided that he would consider any write that provided an annualized total return (capital gains, dividends, and commissions) of at least 12%. This would eliminate many potential writes, but would leave him with a fairly large number of writing candidates each day. He knows the downside break-even point at expiration in each write. Therefore, the probability of the stock being below that break-even point at expiration can be computed quickly. His final list would *rank those writes with the least chance of being below the break-even point at expiration as the best writes*. Again, this ranking is based on an expected probability and is, of course, no guarantee that the stock will not, in reality, fall below the break-even point. However, over time, *a list of this sort should provide the most conservative covered writes*.

Example: XYZ is selling for 43 and a 6-month July 40 call is selling for 8 points. After including dividends and commission costs for a 500-share position, the downside break-even point at expiration is 36. If the annualized volatility of XYZ is 25%, the probability of making money at expiration can be computed. The 6-month volatility is 17.7% (25% times the square root of ½ year). The probability of being below 36 can be computed by using the formula given earlier in this section:

$$P(\text{below 36 in 6 months}) = N\left(\frac{\ln\left(\frac{36}{43}\right)}{.177}\right) = N\left(\frac{-.178}{.177}\right) = 0.158$$

The expected probability of XYZ being below 36 in 6 months is 15.8%. Therefore, this would be an attractive write on a conservative basis, because it has a large probability of making money (nearly 85% chance of not being below the break-even point at expiration). The return if exercised in this example is approximately 20% annualized, so it should be acceptable from a profit potential viewpoint as well. It is a relatively easy matter to perform a similar calculation, with the aid of a computer, on all covered writing candidates.

The ability to measure downside protection in terms of a common denominator—volatility—can be useful in other types of covered call writing analyses. The writer interested in writing out-of-the-money calls, which generally have higher profit potential, is still interested in having an idea of what his downside protection is. He might, for example,

decide that he wants to invest in situations in which the probability of making money is at least 60%. This is not an unusually difficult requirement to fulfill, and will leave many attractive covered writes with a high profit potential to choose from. A downside requirement stated in terms of probability of success removes the necessity of having to impose arbitrary requirements. Typical arbitrary requirements would be including only calls that sell for one point or more, or stating that the downside protection must be a certain percentage of the stock price. These obviously cannot suffice for stocks with different volatilities. Rather, the downside protection criterion should be stated in terms of “probability of down protection” or, alternatively, in terms of the volatility itself. In this manner, a uniform comparison can be made between volatile and nonvolatile stocks.

CALL BUYING

The option buyer can also constructively use the measurement of volatility to aid him in his option buying decisions. In Chapter 3, it was shown that *evaluating the profitability of calls based on the volatility of the underlying stock is the correct way to analyze an option purchase*. One specific method of analysis is described. There are certain variables in this analysis that may be altered to fit the call buyer’s individual preferences, but the general logic is applicable to all cases.

As a first step, *one should decide upon a uniform stock movement for ranking call purchases*. One might decide to rank all purchases by how they would perform if the underlying stock moved up in accordance with its volatility. The phrase “in accordance with its volatility” must be quantified. For example, one might decide to assume that every stock could move up one standard deviation, and then rank all call purchases on that basis. *The prospective call buyer must also fix the time period that he wants to use*. Generally, one looks at purchases to be held for 30 days, 60 days, and 90 days.

The exact steps to be followed in the analysis of profitability and risk can be listed as follows:

1. Specify the distance that underlying stock can move, up or down, in terms of its volatility.
2. Select the holding period over which the analysis is to take place.

PROFITABILITY

3. Calculate the stock price that the stock would move up to, when the foregoing assumptions are implemented.
4. Using a pricing model, such as the Black–Scholes model, estimate what the option price would become after the upward stock movement.

5. Calculate the percent profit, after deducting commissions.
6. Repeat steps 4 and 5 for each option on the stock.

A final ranking of all potential call buys can be obtained by performing steps 3 through 6 on all stocks, and ranking the purchases by their percentage reward.

RISK

7. Calculate the stock price that the stock could fall to, when the assumptions in steps 1 and 2 are applied.
8. With a model, price the option after the stock's decline.
9. Calculate the percentage loss after commissions.
10. Compute a reward/risk ratio: Divide the percentage profit from step 5 by the percentage risk from step 9.
11. Repeat steps 8 through 10 for each option on the stock.

A final ranking of less aggressive option purchases can be constructed by performing steps 7 through 11 on all stocks, and ranking the purchases by their reward/risk ratio.

The higher profitability list of option purchases will tend to be at- or slightly out-of-the-money calls. The less aggressive list, ranked by reward/risk potential, will tend to be in-the-money options.

Example: Steps 1 and 2: Suppose an investor wants to look at option purchases for a 90-day holding period, under the assumption that each stock could move up by one standard deviation in that time. (There is only about a 16% chance that a stock will move more than one standard deviation in one direction in a given time period. Therefore, in actual practice, one might want to use a smaller stock movement in his ranking calculations.) Furthermore, assume that the following data are known:

XYZ common, 41;
XYZ volatility, 30% annually;
XYZ January 40 call, 4; and
time to January expiration, 6 months

Step 3: Calculate upward stock potential. This is accomplished by the following formula:

$$q = pe^{avt}$$

where

p = current stock price

q = potential stock price

v_t = volatility for the time period, t

a = a constant (see below).

The constants, a and t , are fixed under the assumptions in steps 1 and 2. The first constant, a , is the number of standard deviations of movement to be allowed. In our example, $a = 1$. That is, the analysis is being made under the assumption that the stock could move up by one standard deviation. The second constant, t , is .25, since the analysis is for a 90-day holding period, which is 25% of a year. In this example:

$$v_t = v\sqrt{t} = .30 \sqrt{.25} = .30 \times .50 = .15$$

so

$$q = 41e^{.15} = 41 \times 1.16 = 47.64$$

Thus, this stock would move up to approximately 47.64 if it moved one standard deviation in exactly 90 days.

Step 4: Using the Black–Scholes model, the XYZ January 40 call can be priced. It would be worth approximately 8.10 if XYZ were at 47.60 and there were 90 days' less life in the call.

Step 5: Calculate the profit potential. For this example, commissions will be ignored, but they should be included in a real-life situation.

$$\text{Percents profit} = \frac{8.10 - 4}{4} = \frac{4.10}{4} = 103\%$$

Thus, if XYZ stock moves up by one standard deviation over the next 90 days, this call would yield a projected profit of 103%. Recall again that there is only about a 16% chance of the stock actually moving at least this far. If all options on all stocks are ranked under this same assumption, however, a fair comparison of profitable options will be obtained.

Step 6 is omitted from this example. It would consist of performing a similar profit analysis (steps 4 and 5) on all other XYZ options, with the assumption that XYZ is at 47.60 after 90 days.

Step 7: Calculate the downside potential of XYZ. The formula for the downside potential of the stock is nearly the same as that used in step 3 for the upside potential:

$$\begin{aligned}
 q &= pe^{-av_t} \\
 &= 41e^{-.15} = 41 \times .86 = 35.39
 \end{aligned}$$

XYZ would fall to approximately 35.39 in 90 days if it fell by one standard deviation. Note that the actual distances that XYZ could rise and fall are not the same. The upward potential was 6.60 points, while the downward potential is about 5.75 points. This difference is due to the use of the lognormal distribution.

Step 8: Using the Black–Scholes model, one could estimate that the XYZ January 40 call would be worth about 1.10 if XYZ were at 35.39 in 90 days.

Step 9: The risk potential in the January 40 call would be:

$$\text{Percent risk} = \frac{4 - 1.10}{4} = \frac{2.90}{4} = 73\%$$

Step 10: The reward/risk ratio is merely the percentage reward divided by the percentage risk:

$$\text{Reward/risk ratio} = \frac{103\%}{73\%} = 1.41$$

Step 11: This analysis would be repeated for all XYZ options, and then for all other optionable stocks. The less aggressive call purchases would be ranked by their reward/risk ratios, with higher ratios representing more attractive purchases. More aggressive purchases would be ranked by the potential rewards only (step 5).

This completes the call buying example. Before leaving this section, it should be noted that the assumption of ranking the purchases after one full standard deviation movement by the underlying stock is probably excessive. A more moderate assumption would be that the stock might be able to move .7 standard deviation. There is about a 25% expected chance that a stock could move up at least .7 standard deviation at the end of a fixed time period.

PRICING A PUT OPTION

Theoretical models for pricing put options have been derived; that is, ones that are separate from call pricing models. Black and Scholes presented such a model in their original paper. However, as has been demonstrated, there is a relationship between put and call prices in the listed option market due to the conversion and reversal strategies.

One could use the basic call pricing model for the purpose of predicting put prices if he assumes that arbitrageurs will efficiently influence the market via conversions.

Theoreticians will argue that such a method of pricing puts assumes that the arbitrage process is always present and works efficiently, and that this is not true. The “conversion efficiency” assumption could be a serious fault if one were trying to determine the exact overpriced or underpriced nature of the put option. However, if one is merely comparing various put strategies under constant assumptions, the arbitrage model for pricing puts works quite well.

The listed put’s price can be estimated by using the call pricing model and the arbitrage formula. Recall that the arbitrageur must include the cost of carrying the position as well as the dividends to be received.

$$\text{Theoretical put} = \text{Theoretical call price} + \text{Strike price} - \text{Stock price} - \text{Carrying cost} + \text{Dividends}$$

The “theoretical call price” is obtained from the Black–Scholes model. The carrying cost is the cost of money (interest rate) times the striking price, multiplied by the time to expiration. Recall that this is the approximation formula for carrying cost (see Chapter 27 for comments on present value and compounding). Consequently, if XYZ were at 41 and a 6-month January 40 call option were valued at 4 points by the Black–Scholes model, the theoretical put price could be estimated. Assume that the cost of money interest rate is 10% annually, and that the stock will pay \$.50 in dividends in 6 months ($t = \frac{1}{2}$ year).

$$\begin{aligned}\text{Theoretical put price} &= 4 + 40 - 41 - (.10 \times 40 \times \frac{1}{2}) + .50 \\ &= 3 - 2 + .50 \\ &= 1.50\end{aligned}$$

This means that if the call could be sold for 4 points, the arbitrageur would be willing to pay up to 1.50 points for the put to establish a conversion. The arbitrageur’s price is used as the theoretical listed put price estimate.

PUT BUYING

Put option purchases can be ranked in a manner very similar to that described for call option buying. Reward opportunities occur when the stock falls in accordance with its volatility. An upward stock movement represents risk for the put buyer. All of the 11 steps in the previous section on call buying are applicable to put buying. The pricing of the put necessary for steps 4 and 8 is done in accordance with the arbitrage model just presented.

If an underlying stock does not have listed puts trading, the synthetic put can be

considered. While all U.S.-listed stocks have both puts and calls at every strike, there are still situations with warrants, especially in foreign countries, that are applicable to the following discussion. Recall that synthetic puts are created for customers by some brokerage houses. The brokerage sells the stock short and buys a call. The customer can purchase the synthetic put for the amount of the risk involved, plus any dividends to be paid by the underlying stock. The synthetic put pricing formula that would be used in steps 4 and 8 of the option buying analysis is exactly the same as the arbitrage model for listed puts, except that the carrying costs are omitted:

$$\text{Theoretical synthetic put price} = \text{Theoretical call price} + \text{Strike price} - \text{Stock price} + \text{Dividends}$$

When the ranking analysis is performed, very few synthetic puts will appear as attractive put buys. This is because, when the customer buys a synthetic put, he must advance the full cost of the dividend, but receives no offsetting cost reduction for the credit being earned by the short stock position. Consequently, synthetic puts are always more expensive, on a relative basis, than are listed puts. However, if one is particularly bearish on a stock that has no listed puts, a synthetic put may still prove to be a worthwhile investment. The recommended analysis can give him a feeling for the reward and risk potential of the investment.

CALENDAR SPREADS

The pricing model can help in determining which neutral calendar spreads are most attractive. Recall that in a neutral calendar spread, one is selling a near-term call and buying a longer-term call, when the stock is relatively close to the striking price of the calls. The object of the spread is to capture the time decay differential between the two options. The neutral calendar spread is normally closed when the near-term option expires. The pricing model can aid the spreader by estimating what the profit potential of the spread is, as well as helping in the determination of the break-even points of the position at near-term expiration.

To determine the maximum profit potential of the spread, assume that the near-term call expires worthless and use the pricing model to estimate the value of the longer-term call with the stock exactly at the striking price. Since commission costs are relatively large in spread transactions, it would be best to have the computations include commissions. Calculating a second profit potential is sometimes useful as well—the profit if unchanged. To determine how much profit would be made if the stock were unchanged at near-term expiration, assume that the spread is closed with the near-term call equal to its intrinsic

value (zero if the stock is currently below the strike, or the difference between the stock price and the strike if the stock is initially above the strike). Then use the pricing model to estimate the value of the longer-term call, which will then have three or six months of life remaining, with the stock unchanged. The resulting differential between the near-term call's intrinsic value and the estimated value of the longer-term call is an estimate of the price at which the spread could be liquidated. The profit, of course, is that differential minus the current (initial) differential, less commissions.

In the earlier discussion of calendar spreads, it was pointed out that there is both an upside break-even point and a downside break-even point at near-term expiration. These break-even points can be estimated with the use of the pricing model. One method of determination involves estimating the liquidating value of the spread at successive stock prices. When the liquidating value is found to be equal to the initial value, plus commissions, a break-even point has been located.

Example: If the spread in question is using options with a striking price of 30, one would begin his break-even point calculations at a price of 30. Estimate the liquidating value of the spread at 30, 29.90, 29.80, 29.70, and so forth until the break-even point is found. Once the downside break-even point has been determined in this manner, the iterations to locate the upside break-even point should begin again at the striking price. Thus, one would evaluate the liquidating value at 30, 30.10, 30.20, and so on. This is somewhat of a brute-force method, but with a computer it is fairly fast. The number of calculations can be reduced by adopting a more complicated iteration process.

A final useful piece of information can be obtained with the aid of the pricing model—the *theoretical value of the spread*. Recompute the estimated value of both the near-term and longer-term calls at the current time and stock price, using the implied volatility for the underlying stock. The resultant differential between the two estimated call prices may differ substantially from the actual differential, perhaps highlighting an attractive calendar spread situation. One would want to establish spreads in which the theoretical differential is *greater than* the actual differential (that is, he would want to buy a “cheap” calendar spread).

Once these pieces of information have been computed, the strategist can rank the spread possibilities by whatever criterion he finds most workable. *The logical method of ranking the spreads is by their return if unchanged.* The spreads with the highest return if unchanged at near-term expiration are those in which the stock price and striking price were close together initially, a basic requirement of the neutral calendar spread. More complicated ranking systems should try to include the theoretical value of the spread and possibly even the maximum potential of the spread. *A similar analysis can, of course, be worked out for put calendar spreads,* using the arbitrage pricing model for puts.

RATIO STRATEGIES

Ratio strategies involve selling naked options. Therefore, the strategist has potentially large risk, either to the upside or to the downside or both. He should attempt to get a feeling for how probable this risk is. The formulae for determining the probability of a stock being above or below a certain price at some time in the future can give him these probabilities. For example, in a straddle writing situation, the strategist would want to compute such arithmetic quantities as maximum profit potential, return if unchanged, collateral required at upside break-even point or at upside action point (recall that the collateral requirement increases for naked options on an adverse stock movement), and the break-even points themselves. The probabilities of being above the upper break-even point at expiration or below the lower break-even point should be computed as well. Moreover, an expected return analysis could be performed on the position to determine the general level of profitability of the position with respect to all other positions of the same type on other stocks. Such an expected return analysis need not assume that the position is held to expiration. Firm traders, paying little or no commissions, might be interested in seeing the expected results for a holding period as short as 30 days or less. Public customers might use a longer holding period, on the assumption that they would not trade the position as readily because of commission costs. Ratio positions should be ranked either by return if unchanged or by expected return.

The analyses described for calendar spreads and ratio positions should not be relied upon as gospel. In the proposed forms of analysis, one is projecting future option prices and stock prices under the assumption that the volatility of the underlying stock will remain the same. Although this may be true in some cases, there will also be many times when the volatility of the underlying stock will change during the life of the position. If the volatility decreases, the projected break-even points for a calendar spread will be too far away from the striking price. Thus, a loss would result at some prices where the spreader expected to make money. If the volatility increases, the expected return of a ratio position will drop, because the probabilities of the stock moving outside the profit range will increase, thereby increasing the probability of loss.

The effect of a changing volatility can be counteracted, in theory, by continuing to monitor the position daily after it has been established. In a straddle write, for example, if the stock begins to move dramatically, the expected return may become very low. If this happens, adjustments could be made to the position to improve it. Such monitoring is difficult to apply in practice for the public customer, because the commission costs involved in constant position adjustments would mount rapidly. There is no exact method that would allow for infrequent, periodic adjustments, but *by using a follow-up analysis the public customer may be able to get a better feeling for the timing of adjusting a position.* For example, suppose that one initially wrote a 5-point straddle when the stock was

at 30. Sometime later, the stock is at 34. The expected return of writing a 5-point straddle with a strike of 30 when the stock is at 34 could be computed for the shorter time period remaining until expiration. If the expected return is negative, an adjustment needs to be made. Adopting this form of adjusting would keep the number of trades to a minimum, but would still allow the strategist to determine when his position has become improperly balanced. Of course, the current volatility would be used in making these determinations. Another follow-up monitoring technique, using the deltas of the options involved, is presented later in this chapter, and has been described several times previously.

FACILITATION OR INSTITUTIONAL BLOCK POSITIONING

In this and the following section, *the advantages of using the hedge ratio are outlined*. These strategies are primarily member firm, not public customer, strategies, since they are best applied in the absence of commission costs. An institutional block trader may be able to use options to help him in his positioning, particularly when he is trying to help a client in a stock transaction.

Suppose that a block trader wants to make a bid for stock to facilitate a customer's sell order. If he wants some sort of a hedge until he can sell the stock that he buys, and the stock has listed options, he can sell some options to hedge his stock position. To determine the quantity of options to sell, he can use the hedge ratio. The exact formula for the hedge ratio was given earlier in this chapter, in the section on the Black-Scholes pricing model. It is one of the components of the formula. Simply stated, the hedge ratio is merely the delta of the option—that is, the amount by which the option will change in price for small changes in the stock price. By selling the correct number of calls against his stock purchase, the block trader will have a neutral position. This position would, in theory, neither gain nor lose for small changes in the stock price. He is therefore buying himself time until he can unwind the position in the open market.

Example: A trader buys 10,000 shares of XYZ, and a January 30 call is trading with a hedge ratio of .50. To have a neutral position, the trader should sell options against 20,000 shares of stock (10,000 divided by .50 equals 20,000). Thus, he should sell 200 of the January 30's. If the hedge ratio is correct—largely a function of the volatility estimate of the underlying stock—the trader will have greatly eliminated risk or reward on the position for small stock movements. Of course, if the block trader wants to assume some risk, that is a different matter. However, for the purposes of this discussion, the assumption is made that the block trader merely wants to facilitate the trade in the most risk-free manner possible. In this sample position, if the stock moves up by 1 point, the option should move up by 50 cents. The trader would make \$10,000 on his stock position and would lose

\$10,000 on his 200 short options—he has no gain or loss. Once the trader has the neutral position established, he can then begin to concentrate on unwinding the position.

In actual practice, this hedge ratio may not work exactly, because it tends to change constantly as the stock price changes. If the trader finds the stock moving more than fractionally, he may have to add more calls or buy some in, to maintain a neutral hedge ratio. This would expose him to some risk, but the risk is substantially smaller than if he had not hedged at all. Of course, there would also be certain cases in which he would profit by the stock price change. For example, implied volatility could decrease, making the calls cheaper.

A similar hedge can be established by the block trader who sells stock to accommodate a customer buy order. He could buy calls in accordance with the hedge ratio, to set up a neutral position.

This process of facilitation is quite widely practiced, especially by brokerage houses that are trying to attract the business of the large institutional customer. Since the introduction of listed call options and their applications for facilitating orders, many quotes for large blocks of stock have improved considerably. The block trader (who works for the brokerage house) is willing to make a higher bid or a lower offer if he can use options to hedge his position. This facilitation with options results in a better market (higher bid or lower offer) from the point of view of the institutional customer. Without the availability of such listed options, the block trader would probably make a bid or an offer that was substantially away from the prevailing market price in order to work out of his stock-only position with a lessened degree of risk. This would obviously present a poorer market for the institutional customer.

THE NEUTRAL SPREAD

The hedge ratios (deltas) of two or more options may be used to determine a neutral spread. This strategy is especially useful to market-makers on the options exchanges who may want to reduce the risk of options bought or sold in the process of providing a public market. If the hedge ratios of two options are known, *the neutral ratio is determined by dividing the two hedge ratios.*

Example: An XYZ January 35 has a hedge ratio of .25, and an XYZ January 30 has a hedge ratio of .50, so a neutral ratio would be 2:1 (.50 divided by .25). That is, one would sell 2 January 35's against one long January 30, or, conversely, would buy 2 January 35's against one short January 30. Thus, a market-maker who has just bought 50 January 30's in an effort to provide a market for a public seller of that call could hedge his position by selling 100 January 35's. This should keep his risk small, for small stock price changes,

until he can unwind the position. The ratio for the neutral spread is not as sensitive to the volatility estimate of the underlying stock as is the ratio concerning stock and options. This is because the same volatility estimate is applied to both options, and the resultant ratio for the spread would not tend to change greatly.

The risk trader can also use the neutral spread ratio to his advantage. This concept was illustrated several times in previous chapters describing ratio writing, ratio spreads, and straddle writes. Ratio spreads are quite popular with member firm traders and floor traders. Recall that a ratio spread consists of buying options at a certain strike, and selling more options further out-of-the-money. The hedge ratios can, of course, be used by the trader, or by a public customer, to initially establish a neutral position. Perhaps more important, the hedge ratio can also be used as a follow up action to keep the position neutral after the stock changes in price. This strategy is the “delta spread” described in Chapter 11.

The risk trader is not attempting to establish the spread with the idea of minimizing risk for small stock movements. Rather, he is looking to make a profit, but would prefer to remain as neutral as possible on the underlying stock. He is implementing a risk strategy that has a neutral outlook on the underlying stock. He is selling much more time value premium than he is buying.

Example: The purchase of 15 January 30 calls and the sale of 30 January 35 calls—a ratio call spread—may be a position taken for profit potential. It would be a neutral position if the deltas were .60 and .30, for example. This spread would do best if the stock were at exactly 35 at expiration. However, if the stock rose quickly before expiration, the spread ratio would decrease from 2:1 to perhaps 3:2. That is, the neutral ratio between the January 30 call and the January 35 call should be 3 short January 35's to 2 long January 30's. If the trader wants to balance his position, he could buy 5 more January 30's, giving him a total of 20 long versus the 30 short January 35's that he originally sold. Conversely, if the stock dropped in price, the neutral spread ratio might increase, indicating that more calls should be sold. For example, if this stock declines, the neutral ratio might be 3:1. In that case, 15 more January 35's could be sold, making the position short 45 calls versus 15 long calls, which would produce the neutral 3:1 ratio.

It would not be proper to adjust the ratio constantly, because the frequent whipsaw losses on trading movements would wipe out the profit potential of the position. However, the trader may want to pick out points, in advance, at which he wants to reevaluate his position before something drastic goes wrong. For example, if the foregoing spread were established with the stock at a price of 30, the spreader might want to readjust at 33 or 27, whichever comes first.

By monitoring the spread using the hedge ratio, the trader may also be able to discern whether he has established too bullish or too bearish a position.

Example: The trader starts with the example described above—long 15 January 30 calls and short 30 January 35 calls—when the hedge ratios were .60 and .30, respectively. Some later time, the stock falls to 27 and the trader needs to reevaluate his position. The hedge ratios may have become .42 for the January 30 and .14 for the January 35, indicating that a 3:1 ratio would be neutral ($.42/.14 = 3$). He now has a bullish position, because his 2:1 ratio is *less than* the neutral 3:1 ratio. It is not mandatory that the trader act on this information. He may actually be bullish on the stock at this point and decide to remain with his position. The usefulness of the hedge ratio is that it allows him to see that his position is bullish, so he can make a correct judgment. Without this knowledge, he might still think his position to be neutral, a critical mistake if he indeed wants to be neutral. If the trader's ratio is *greater than* the neutral ratio (2:1 vs. 3:2, for example), he is bearishly positioned.

As a final point, it should be noted that the ratio can be adjusted by buying or selling either option.

Example: If the stock falls and it is desired that the ratio be increased to 3:1, one might sell more January 35's or might decide to sell out some of his January 30's. A bullish adjustment could be made by buying on either side of the spread in a similar manner. In general, one should adjust by selling time premium or buying intrinsic value. That is, out-of-the-moneys are usually sold and in-the-moneys are usually bought, when adjusting.

AIDING IN FOLLOW-UP ACTION

The computer can also be an invaluable aid to the strategist in that it can help him monitor his positions. It is generally necessary for the strategist to have some way of inputting his positions into an inventory database and also to have some way of identifying different securities that are grouped within the same trading position. Once this has been done, *the computer can simultaneously read pricing data (either realtime or closing prices) and the inventory database to generate information concerning the current status of any position.*

A current mark to market (profit and loss) statement is of obvious use in that the trader can see how he is doing each day. The computer can also easily generate a set of warning flags that may be of interest to the trader, and could produce a list summarizing possible positions that need action. In most of the strategies that were described, it was shown that the strategist should avoid early assignment if at all possible. It is a simple matter for the computer to calculate the remaining time value premium of any short options, and to warn the trader if there is only a small amount of time value premium remaining, perhaps 0.10 or less. For similar reasons, the trader may want to have a daily list of positions

that are nearing maturity, perhaps with less than 1 month of life remaining in the options. A flag indicating an approaching ex-dividend date might also be useful for this purpose.

If the trader inputs another piece of information into the database, the computer can help him in another follow-up action. In most strategies that were described, especially those involving uncovered options, the trader wants to take some sort of follow-up action based on the price movement of the underlying stock. If the stock rallies too far, he may want to cover short calls or buy other calls as protection. If the stock declines too far, similar maneuvers would apply to put options or to rolling down short calls. If the trader inputs the stock prices at which he would like to take action, the computer can monitor each day's closing price of the stock and generate a list of positions that have exceeded their upside or downside action points.

The computer can also do more sophisticated types of position monitoring. Recall that it was pointed out that the deltas of the options involved in a position can be compared to each other to tell whether the position is bullish or bearish. The Black-Scholes model can be used to calculate the deltas of the options in one's positions. Then the net position can be determined by the computer, thereby telling the trader whether his position has become "delta long" (bullish), "delta short" (bearish), or neutral. If he sees that a position is bearish and he does not want to be structured in that way, he can make bullish adjustments. The delta spread and neutral spread strategies very conveniently lend themselves to such types of follow-up action, although any of the more complicated straddle writing and protected straddle writing positions can be monitored usefully in this way as well.

The computation for determining whether a position is net short or net long generally involves calculating the "equivalent stock position" (ESP). If one owns 10 calls that have a delta of .45, his equivalent stock position from those calls is $10 \times 100 \text{ shares per call} \times .45 = 450$. That is, owning those 10 calls is equivalent to owning 450 shares of the underlying stock, according to the delta. All puts and calls can be reduced to an ESP and can then, of course, be combined with any actual long or short stock in the position to produce an ESP for the entire strategy. The resultant ESP for each of the trader's positions can be printed from the computer along with the items described above.

Further sophisticated measures can be taken. The computer can generate a table of results at expiration. If so desired, this could be presented as a graph, but that is not really necessary. A table suffices quite well, as shown by most of the examples in this book. Such a picture has meaning only if all options in the position expire at the same time. If they don't, one may instead want the computer to compose a table of results or a graph at *near-term expiration*. Thus, in a calendar spread, for example, one could see what sorts of profitability he would be looking at when it was time to remove the spread.

Finally, the computer can compute the expected return of a position already in place. This would give a more dynamic picture of the position, and this expected return

is usually for a relatively short time period. That time period might be 30 days, or the time remaining until expiration, whichever is less. The expected return is calculated in much the same manner as the expected return computation described earlier in this chapter. First, one uses the stock's volatility to construct a range of prices over which to examine the position. Second, one uses the Black–Scholes model to calculate the values of the various options in the position at that future time and at the various stock prices. Some of the results should be displayed in table form by the computer program. The expected profit is computed, as described earlier, by summing the multiples of the probabilities of the stock being at each price by the result of the position at that price. The expected return is then computed by dividing the expected profit by the expected investment. Since margin computations can require involved computer programs, it is sufficient to omit this last step and merely observe the expected profit. The following example shows how a sample position might look as the computer displays the position itself, the ESP, the profit at expiration, and the expected profit in 30 days. A complex position is assumed, in order that the value of these analyses can be demonstrated.

Example: The following position exists when XYZ is at 31.75. It is essentially a backspread combined with a reverse ratio write. It resembles a long straddle in that there is increased profit potential in either direction if the stock moves far enough by expiration. Initially, the computer should display the position and the ESP.

Position		Delta	ESP	
Short	4,500 XYZ	1.00	Short	4,500 shares
Short	100 XYZ April 25 calls	0.89	Short	8,900 shares
Long	50 XYZ April 30 calls	0.76	Long	3,800 shares
Long	139 XYZ July 30 calls	0.74	Long	10,286 shares
Total ESP			Long	686 shares
Total money in position: \$163,500 credit				

The advantage of using the ESP is that this fairly complex position is reduced to a single number. The entire position is equivalent to being long 686 shares of the common stock. Essentially, this is close to delta-neutral for such a large position. The next item that the computer should display is the total credit or debit in the position to date. With this information, an expiration picture can be drawn if it is applicable. In this position, since there is a mixture of April and July options, a strict expiration picture does not apply. Rather, the computer should draw a picture based on the position at April expiration or on a shorter time horizon.

Assume that April expiration is still some time away, so that the computer will

instead draw the picture 30 days hence. In order to do so, the computer uses the stock's volatility to project stock prices 30 days in the future. Seven stock prices are shown in the next table; they represent points along the distribution curve of the stock, ranging from minus one and one-half standard deviations to plus one and one-half standard deviations of movement from the current stock price. While these seven points certainly do not comprise the entire spectrum of possible stock moves, they are a representative sample.

Stock Price in 30 Days	Standard Deviations	Expected Results
35.90	+1.5	+\$15,847
34.10	+1.0	+ 12,355
32.90	+0.5	+ 10,097
31.75	0.0	+ 9,443
30.60	-0.5	+ 10,743
29.50	-1.0	+ 14,172
28.50	-1.5	+ 19,605
Expected profit		+\$11,426

Obviously, this position has had some profitable adjustments made to it in the past. That is not important at this point, because the trader is interested only in the future. If the current mark to market of this position were in excess of \$11,426, then he should consider removing the position, since it would be more profitable than the expected profit.

IMPLEMENTATION

Many of the analyses described in this chapter can be obtained from a reliable data service or brokerage firm. The strategist who plans to prepare his own analysis, either by himself or by contracting the programming work out, should be aware that computer programs should not be written in website languages such as Java Script, PHP, HTML, etc., because the mathematics are far too complicated. Languages such as Pascal, C, C++, Visual Basic, or any high-level structured programming language would suffice, although Java could be considered as well, keeping in mind that Java's main usage is not for computational purposes. Reliable option pricing data that include dividend information on the underlying stock are also necessary. The larger programmable calculators can handle calculations such as the Black-Scholes model, computing the hedge ratio, and determining the probability of a stock being above or below a certain price at some future time.

However, more involved calculations, such as computing the implied volatility or determining the expected return of a position, require the use of a computer.

SUMMARY

Two basic mathematical aids have been presented: the pricing model and the ability to predict the probability of a stock's movement. The hedge ratio and the expected return analysis are extensions of the basic aids. Any strategy can be evaluated with these tools. Such an analysis should be able to give the trader or strategist some idea of the relative attractiveness of establishing the position, and may also aid in making follow-up adjustments to the position. All the analyses rely heavily on one's estimate of the volatility of the underlying stock. Using the implied volatility seems to be one of the best ways to obtain an accurate, current volatility estimate, since it is derived from the prices in the market itself. The applications presented here are not all-inclusive. The strategist who is, or becomes, familiar with the advantages of rigorous mathematical analysis will be able to construct many other aids for his trading that utilize the basic mathematics described in this chapter.